

Yichuan Wang

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EDUCATION

University of California, Berkeley

Ph.D. in EECS, Sky Computing Lab

- Advised by [Matei Zaharia](#) & [Joseph E. Gonzalez](#)
- Closely collaborating with [Sewon Min](#)

Berkeley, California

Aug. 2024 - Present

Shanghai Jiao Tong University

B.Eng. in Computer Science, ACM Honors Class

Shanghai, China

Sept. 2020 - Jun. 2024

RESEARCH INTERESTS

Machine Learning Systems; Large Language Models (LLMs) & Foundation Models; Agentic AI Systems; Scalable Vector Databases & Retrieval-Augmented Generation (RAG); Post-Training & Inference Optimization; Multi-Modal Embeddings & Semantic Search; Efficient Model Serving & Scheduling; GNN at Scale.

EXPERIENCE

Shanghai Jiao Tong University

Undergraduate Researcher in Machine Learning Systems, advised by Prof. [Quan Chen](#)

Shanghai, China

June 2022 - June 2024

New York University

Research Assistant in Machine Learning Systems, advised by Prof. [Jinyang Li](#)

New York, NY, USA

Jan. 2023 - Apr. 2024

LMSYS (SGLang Project)

Major Developer

Remote

Jul. 2024 - Dec. 2024

Meta (MSL Infra, PyTorch)

Research Scientist Intern, Post-Training Infrastructure

Menlo Park, CA, USA

May 2026 - Present

SELECTED OPEN-SOURCE PROJECTS

LEANN – Low-Storage Vector Index for Personal RAG ([GitHub](#))

Aug. 2025 – Present

- Creator and lead maintainer of LEANN, a storage-efficient vector index for on-device RAG that reduces index size to under 5% of raw data while maintaining high recall and efficiency.
- Popular open-source repository to apply research project to deploy RAG and agent on edge devices; adopted by thousands of developers; the repository has **12k** GitHub stars with **30+** active external contributors and **100k+** downloads.

PUBLICATIONS

LEANN: A Low-Storage Vector Index

Yichuan Wang, Shu Liu, Zhifei Li, Yongji Wu, Ziming Mao, Yilong Zhao, Xiao Yan, Zhiying Xu, Yang Zhou, Ion Stoica, Sewon Min, Matei Zaharia, Joseph E. Gonzalez

- Best Paper Award at MLSys 2026** (short version in VecDB@ICML 2025) [[pdf](#)]
- Adopted by thousands of developers; **12k** GitHub stars, **30+** active external contributors, **100k+** downloads [[GitHub](#)]

PixelRAG: Visually Grounded Retrieval-Augmented Generation with Screenshot Rendering

Yichuan Wang*, Zhifei Li*, Zirui Wang, Paul Teilletche, Lesheng Jin, Matei Zaharia[†], Joseph E. Gonzalez[†], Sewon Min[†]

- * indicates equal contribution, [†] indicates equal advising
- Preprint [[pdf](#)] [[GitHub](#)]

DS SERVE: A Framework for Efficient and Scalable Neural Retrieval

Jinjian Liu*, **Yichuan Wang***, Xinxi Lyu, Rulin Shao, Joseph E. Gonzalez, Matei Zaharia, Sewon Min

- * indicates equal contribution
- Accepted by **AAAI 2026** (Demo) [[pdf](#)]
- Deployed the largest public vectorstore over pretraining data; serving **1k+ queries/day** [[Demo](#)] [[Blog](#)]

Locality-aware Fair Scheduling in LLM Serving

Shiyi Cao*, **Yichuan Wang***, Ziming Mao, Pin-Lun Hsu, Liangsheng Yin, Tian Xia, Dacheng Li, Shu Liu, Yineng Zhang, Yang Zhou, Ying Sheng, Joseph E. Gonzalez, Ion Stoica

- * indicates equal contribution
- Preprint [\[pdf\]](#)
- Production deployment: integrated into **Yandex's** LLM serving infrastructure and **SGLang's** open-source serving framework

DiskGNN: Bridging I/O Efficiency and Model Accuracy for Out-of-Core GNN Training

Renjie Liu*, **Yichuan Wang***, Xiao Yan, Zhenkun Cai, Minjie Wang, Haitian Jiang, Bo Tang, Jinyang Li

- * indicates equal contribution
- Accepted by **SIGMOD 2025** (Oral) [\[pdf\]](#)
- Collaborated with AWS AI Lab

Optimizing Dynamic Neural Networks with Brainstorm

Weihaio Cui, Zhenhua Han, Lingji Ouyang, **Yichuan Wang**, Ningxin Zheng, Lingxiao Ma, Yuqing Yang, Fan Yang, Jilong Xue, Lili Qiu, Lidong Zhou, Quan Chen, Haisheng Tan, Minyi Guo

- Accepted by **OSDI 2023** [\[pdf\]](#)
- Collaborated with MSRA

Forming Scalable, Convergent GNN Layers that Minimize a Sampling-Based Energy

Haitian Jiang, Renjie Liu, Zengfeng Huang, **Yichuan Wang**, Xiao Yan, Zhenkun Cai, Minjie Wang, David Wipf

- Accepted by **ICLR 2025** [\[pdf\]](#)

The Danger of Overthinking: Examining the Reasoning-Action Dilemma in Agentic Tasks

Alejandro Cuadron, Dacheng Li, Wenjie Ma, Xingyao Wang, **Yichuan Wang**, Siyuan Zhuang, Shu Liu, Luis Gaspar Schroeder, Tian Xia, Huanzhi Mao, Nicholas Thumiger, Aditya Desai, Ion Stoica, Ana Klimovic, Graham Neubig, Joseph E. Gonzalez

- Preprint [\[pdf\]](#)

Autellix: An Efficient Serving Engine for LLM Agents as General Programs

Michael Luo, Xiaoxiang Shi, Colin Cai, Tianjun Zhang, Justin Wong, **Yichuan Wang**, Chi Wang, Yanping Huang, Zhifeng Chen, Joseph E. Gonzalez, Ion Stoica

- Accepted by **NSDI 2026** [\[pdf\]](#)

INVITED TALKS

- [10/2025] **Lightspeed interview** – LEANN: Towards Lightweight Vector Search and RAG Everything on PC
- [09/2025] **Bytedance Data Team** – LEANN: Towards Lightweight Vector Search and RAG Everything on PC
- [07/2025] **SIGMOD 2025** – DiskGNN: Bridging I/O Efficiency and Model Accuracy for Out-of-Core GNN Training
- [01/2025] **LMsys** – SGLang-FLPM [\[Video\]](#)

SERVICES

- SIGMOD'25 Artifact Evaluation Committee (04/2025)
- MLSys'25 Artifact Evaluation Committee (03/2025)
- EuroSys'25 Artifact Evaluation Committee (02/2025)
- SIGCOMM'24 Artifact Evaluation Committee (07/2024)
- OSDI'24 Artifact Evaluation Committee (04/2024)
- USENIX ATC'24 Artifact Evaluation Committee (04/2024)

TECHNICAL SKILLS

Programming Languages: Python, CUDA, Triton, Java, Rust, Verilog

Reference: [GitHub Profile](#)